

Programme	BSc Mathematics Honours			
Course Code	MAT4CJ205			
Course Title	FUNDAMENTALS OF PYTHON AND SAGEMATH			
Type of Course	Major			
Semester	IV			
Academic Level	200-299			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	3	2	75
Pre-requisites	1) Basic knowledge to start a desktop/laptop computer 2) A basic course in calculus with an understanding of differential and integral calculus (higher secondary level and one or two semester courses from Bsc) 3) A basic course in linear algebra ((higher secondary level))			
Course Summary	In the first part of the course, it intends to give a quick introduction to writing python programs using various popular interfaces. How to handle data and save and read them files is introduced next along with the concepts of repeating the tasks using conditionals and loops. The problems connected with matrices and arrays is solved using the python module numpy. The python module SymPy is used to do various mathematical problems related with symbolic computations. A brief introduction of python module pandas is given, which is used to do data analysis. Using the Python programming structure, an introduction to the advance mathematics software sagemath is given in the second part of the course. Various practical problems making use of concepts from the calculus and linear algebra are to be solved using the sagemath software so that the students will come to know some of the applications of mathematics in real life.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category #	Evaluation Tools used
CO1	Develop proficiency in fundamental to advanced Python programming concepts, including variables, data types, control structures, functions, modules, file handling, and matrix operations.	C	C	Internal Exam/Quiz/End Sem
CO2	Demonstrate competence in data visualization techniques using Matplotlib, encompassing plotting mathematical functions, 2D and 3D graphics, and animated plots.	Ap	C	Internal Exam /Assignment/ End Sem
CO3	Develop proficiency in symbolic computation with SymPy, data manipulation with Pandas, and algebraic computations with SageMath, enabling them to solve diverse mathematical problems numerically and analytically.	C	C	Internal Exam /viva/ Seminar/End Sem
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook	1. Ajith Kumar B.P., Python for Education, https://scischool.in/python/pythonForEducation.pdf 2. Gregory V. Bard, Sage for Undergraduates (online version) http://www.people.vcu.edu/~clarson/bard-sage-for-undergraduates-2014.pdf 3. Tuan A. Le and Hieu D. Nguyen, SageMath Advice For Calculus, https://users.rowan.edu/~nguyen/sage/SageMathAdviceforCalculus.pdf		
Module	Unit	Content	Hrs (45+ 30)
I	Introductory Python and Arrays (Text 1: Chapter 2, Chapter 3)		
	1	Section 2.1: Getting started with Python Section 2.2: Variables and Data Types, Keywords, Section 2.3: Operators and their Precedence.	12
	2	Section 2.4: Python Strings Section 2.5: Python Lists Section 2.6: Mutable and Immutable Types. Section 2.7: Input from the Keyboard Section 2.8: Python Syntax, Colon & Indentation	
	3	Section 2.9: Controlling the Programe Flow Section 2.10: Iteration: for loops Section 2.11: Conditional Execution: if, elif and else Section 2.12: Modify loops: break and continue..	
	4	Section 2.15: Functions Section 2.17: Python Modules and Packages. Section 2.18: File Input/Output Section 2.19: Formatted Printing. Section 2.21: Matrices in pure Python.	
	5	All topics up to Section 3.1, Section: 3.1: NumPy Arrays	
	6	Section: 3.2: Vectorizing Functions.	
II	Data Visualization (Text 1: Chapter 4)		

	7	Section: 4.1: The Matplotlib Module	10
	8	Section: 4.2: Plotting mathematical functions Section: 4.3: Plotting Error Bars, Section: 4.4: Simple 2D animation.	
	9	Section: 4.5: Famous Curves Section: 4.6: 2D plot using colors.	
	10	Section: 4.7: 3D Plots.	
III	Introduction to SymPy and Pandas (Text 1: Chapter 5 and Chapter 6)		
	11	All topics up to Section 5.1, Section 5.1: SymPy, Symbolic Computation in Python.	10
	12	Section 5.2: SymPy, Derivative and Integral	
	13	Section 5.3: SymPy, Operation on sets	
	14	Section 6.1: Series	
	15	Section 6.2: Data Frame	
	16	Section 6.3: Practical Examples	
IV	Sagemath – An Introduction		
		(Text 2: Chapter 1, For units 17,18,19)	
	17	Getting and installing sagemath in Windows, Ubuntu OS Using sagemath using cocalc (online) Section 1.1: Using Sage as a Calculator Section 1.2: Using Sage with Common Functions Section 1.3 : Using Sage for Trigonometry	13
	18	Section 1.5: Matrices and Sage, Part One 1.5.1: A First Taste of Matrices 1.5.3: Doing the RREF in Sage	
	19	Section 1.5: Using Sage to Manipulate Polynomials	
		(Text 3: Chapter 2, 3, 5, For units 20,21,22)	
	20	Section 2.1: Plotting Graphs	
	21	Section 3.1: The Derivative Section 3.2: Higher-Order Derivatives	

	22	Section 5.1: Antiderivatives (Indefinite Integral), Section 5.2: Riemann Sums and the Definite Integral All topics up to 5.2.1, 5.2.1: Riemann Sum Using Left Endpoints	
		Practical (Open-ended)	
		Online References for Practical	30
		1 Python official website and documentation, https://www.python.org/ 2 Spyder official website and documentation, https://www.spyder-ide.org/ 3 Getting Started: Python and IDLE, MIT Courseware, https://web.mit.edu/6.s189/www/handouts/GettingStarted.html 4 Jupyter Notebook, https://jupyter.org/ 5 Google Colaboratory (colab), https://colab.google/ 6 Pydroid 3 IDE for Android (https://play.google.com/store/apps/details?id=ru.iiec.pydroid3&hl=en_US&pli=1) with Pydroid 3 repository plugin (https://play.google.com/store/apps/details?id=ru.iiec.pydroid3.quickinstallrepo&gl=US).	
		Practical problems in basic Python	
		1) Write a programme to work as a basic Income Tax Calculator 2) Write a program that takes the length of an edge (an integer) as input and prints the cube's surface area as output. 3) Write a loop that counts the number of space characters in a string. Recall that the space character is represented as ' '. 4) Write a while loop that computes the factorial of a given integer N.	

	5) Write a program that computes square roots. 6) Write a programme for data Encryption based on Caesar shift. 7) Develop a program that computes the Flesch Index for a text file. 8) Using a List to Find the Median of a Set of Numbers 9) Finding the Mode of a List of Values. Numerical methods using python (Text1: Chapter 7)(7.1 - 7.10, 7.12) 1) Evaluate a Taylor series numerically. 2) Interpolate a function using a) Newton's forward interpolation b) Newton's backward interpolation c) Lagrange's Interpolation d) Newton's General Interpolation 3) Find integral of function using a) Trapezoidal rule b) Simpson's 1/3-rule 4) Find derivative of function numerically. 5) Solve first order differential equations numerically. a) Euler method b) Fourth order Runge-Kutta method 6) Solve algebraic equations numerically. a) The Bisection method b) Regula Falsi Method Practical problems using numpy, matplotlib, pandas and sympy 1) Various vector operations. such as dot product, cross product and divergent using numpy module. 2) Various matrix operations such as determinant, inverse and transpose using numpy module. 3) Solve system of linear equations using numpy module. 4) Plot various 2-D, 3-D curves using matplotlib module.
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	5) Plot various 3-D surfaces using matplotlib module. 6) Find maxima and minima of a function using SymPy module. 7) Necessary data analysis of a given data using pandas module. Practical problems in Sage 1) Solve a system of linear equations (Text 2) 2) Constrained Optimization by Lagrange Multipliers (Text 2, 4.18.2) 3) Traffic Flow (Text 3) 4) Minimum Cost (Text 3) 5) Packaging (Minimum Surface Area) (Text 3) 6) Maximize Revenue (Text 3) 7) Area Between Curves (Text 3) 8) Average Value and mean value theorem (Text 3, 6.2) 9) Newton's Method to find approximate roots (Text 3)	
References: 1 Amit Saha, Doing Math with Python, No Starch Press, 2015. 2 Vernon L. Ceder, The Quick Python Book, Second Edition, Manning. 3 Python tutorial online, https://www.geeksforgeeks.org/python-programming-language/ 4 2D plotting, https://doc.sagemath.org/html/en/reference/plotting/sage/plot/plot.html 5 3D Graphics, https://doc.sagemath.org/html/en/reference/plot3d/index.html 6 Linear Algebra, https://doc.sagemath.org/html/en/tutorial/tour_linalg.html 7 John Harris, Karen Kohl, and John Perry, Peering into Advanced Mathematics through Sage-colored Glasses 8 Paul Zimmermann, Alexandre Casamayou, Computational Mathematics with SageMath, https://www.sagemath.org/sagebook/english.html Kenneth A Lambert, Fundamentals of Python First Programs, Edn 2, Cengage		

***70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	2	3	1	3	2	3	3	1	1	2
CO 2	2	2	3	1	3	2	3	3	1	1	2
CO 3	2	2	3	1	3	2	3	3	1	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Quiz
- Practical Based Assessment
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Quiz	Viva	Practical based assessment	End Semester Examinations
CO 1	√			√		√	√
CO 2	√	√				√	√
CO 3	√		√		√	√	√